Thm. Let f be an analytic on the closed disk $|z-z_0| \le r$. Then

$$f(z_0) = \frac{1}{2\pi} \cdot \int_0^{2\pi} f(z_0 + r \cdot e^{i\theta})\, d\theta.$$

Proof. By the Cauchy's integral formula,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_{|z-z_0|=r} \frac{f(z)}{z - z_0}\, dz$$

Then, for a parameterization $z(\theta) = z_0 + r \cdot e^{i\theta}$ $(0 \le \theta \le 2\pi)$,

$$= \frac{1}{2\pi i} \cdot \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{re^{i\theta}} \cdot r \cdot i e^{i\theta}\, d\theta$$

$$= \int_0^{2\pi} f(z_0 + re^{i\theta})\, d\theta \qquad \blacksquare$$

This theorem is called the Gauss's Mean-value Theorem.

Maximum Modulus Theorem.

If non-constant f is analytic in a domain D, then $|f|$ cannot attain a maximum in D.

Proof. Suppose that $|f|$ attains maximum at $z_0 \in D$. (Then f must be a constant.)

Since D is open, for some $r > 0$, $\overline{B_r(z_0)} \subseteq D$. Then, the Gauss mean-value Thm gives

$$|f(z_0)| \le \frac{1}{2\pi} \cdot \int_0^{2\pi} |f(z_0 + re^{i\theta})|\, d\theta \le \frac{1}{2\pi} \cdot \int_0^{2\pi} |f(z_0)|\, d\theta = |f(z_0)|$$

$|f(z_0)|$ is maximum.

So that $|f(z_0)| = \frac{1}{2\pi} \cdot \int_0^{2\pi} |f(z_0 + re^{i\theta})|\, d\theta$, or

$$\frac{1}{2\pi} \cdot \int_0^{2\pi} |f(z_0)| - |f(z_0 + re^{i\theta})|\, d\theta = 0.$$

By the assumption, the term in the integral is non-negative, thus

$$|f(z_0 + re^{i\theta})| = |f(z_0)| \quad \text{for all } 0 \le \theta \le 2\pi.$$

for all $0 < \varepsilon \le r$, $|f(z_0)| = |f(z_0 + \varepsilon \cdot e^{i\theta})|$ and so $|f(z)| = |f(z_0)|$ for all $z \in \overline{B_r(z_0)}$.

So, f is constant, the identity theorem gives the result.